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Sparse Prototype Code Underlies Classification and Prediction Across Modalities

A Universal Geometry of Within-Class Variability, Aligned With Class Centroids and Competing Classes, Emerges Across Vision, Audio, and Language Models

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Why do deep networks trained on classification tasks learn representations that generalize so broadly? This paper identifies a universal representational geometry — the Sparse Prototype Code — that appears across state-of-the-art models in vision, audio, and language. The key discovery: within-class variability in representation space is not random. Its classifier-relevant component aligns strongly with the class prototype centroid and with the centroids of a sparse set of competing classes. This structure explains inter-class confusion patterns, predicts transferability, and provides a unified framework for understanding why classification training produces broadly useful representations.

AT A GLANCE

Problem: The geometric structure of learned representations is poorly understood, limiting principled model design and analysis.

Why it matters: A universal geometry across modalities would explain why classification training generalizes and enable principled comparison of representations.

Method: Spectral analysis of within-class covariance matrices in final representation spaces of vision, audio, and language models.

Result: Sparse Prototype Code geometry confirmed across all tested architectures and modalities; predicts confusion patterns and transferability.

Evidence: 33 pages, 13 figures, 14 tables spanning ViT, ResNet, CLIP, wav2vec, BERT, and GPT-class models.

The Big Picture

Deep learning models trained on classification tasks have proven surprisingly versatile: representations learned to classify ImageNet transfer to medical imaging, satellite imagery, and fine-grained recognition; representations learned on language classification transfer to generation, translation, and reasoning. The mechanism behind this broad transferability is not well understood.

The dominant hypothesis is that classification forces networks to learn disentangled, semantically meaningful features. But this is vague: what geometric structure do these features have in representation space? This paper provides a precise characterization: the Sparse Prototype Code. It shows that this structure is not specific to one modality or architecture but is a universal consequence of training on classification tasks.

Why Existing Approaches Fall Short

Neural collapse theory (Papyan et al., 2020) characterizes the geometry of the final layer at the end of training: class means collapse to the vertices of a simplex equiangular tight frame (ETF). This is a useful characterization of inter-class geometry but does not address *within-class* variability — the spread of examples around each class mean. Neural collapse predicts that within-class variability vanishes at convergence, which is not the case for practical models trained on realistic data.

CKA and representation similarity methods compare representations between models but do not characterize the intrinsic geometry of a single model's representation space.

Prototype theory in cognitive science posits that categories are organized around prototypical examples, but does not make quantitative predictions about the geometric structure of computational model representations.

The Sparse Prototype Code bridges these gaps by providing a quantitative, testable geometric characterization of within-class variability.

Core Mechanism

For a model f_θ and class c, let $\mu_c = \mathbb{E}_{x\sim p(x|c)}[f_\theta(x)]$ be the class centroid and $\Sigma_c = \mathbb{E}_{x\sim p(x|c)}[(f_\theta(x) - \mu_c)(f_\theta(x) - \mu_c)^\top]$ be the within-class covariance.

The paper performs spectral analysis of Σ_c, decomposing it as $\Sigma_c = U_c \Lambda_c U_c^\top$. The key finding is that the leading eigenvectors $u_1, u_2, \dots$ of Σ_c align strongly with class centroids:

1. $u_1 \approx \mu_c/\|\mu_c\|$ (the class prototype direction)
2. $u_2, \dots, u_k \approx \mu_{c_j}/\|\mu_{c_j}\|$ for a sparse set of competing classes $\{c_j\}_{j=1}^k$, with $k \ll C$ (where C is the total number of classes)

The eigenvalue spectrum is sparse: a small number of eigenvalues are large (aligned with prototype and competing-class directions) and the remaining eigenvalues are small (random variability). This sparsity is the defining feature of the Sparse Prototype Code.

Technical Formulation

Formally, define the *prototype alignment* of class c:

$$A_{\text{proto}}(c) = \frac{u_1^\top \mu_c}{\|\mu_c\|}$$

and the *competing-class alignment* of class c with class c':

$$A_{\text{comp}}(c, c') = \max_{j \geq 2} \frac{u_j^\top \mu_{c'}}{\|\mu_{c'}\|}$$

The Sparse Prototype Code hypothesis states:

$$A_{\text{proto}}(c) \approx 1 \quad \text{for all } c$$

$$\sum_{c' \neq c} \mathbf{1}[A_{\text{comp}}(c, c') > \tau] = k(c) \ll C$$

where $k(c)$ is a sparse count of strongly competing classes. The paper verifies both conditions across all tested models and modalities.

Learning or Inference Procedure

The analysis is purely empirical and requires no training. For each model:

1. Extract final-layer representations for all examples in the test set.
2. Compute class centroids $\{\mu_c\}$.
3. Compute within-class covariance matrices $\{\Sigma_c\}$.
4. Perform eigendecomposition of each Σ_c.
5. Measure alignment of leading eigenvectors with class centroids.

All steps are straightforward linear algebra; no optimization is involved.

What the Guarantee Says

The paper provides a theoretical derivation showing that the Sparse Prototype Code is the *optimal* within-class geometry for a linear classifier that minimizes cross-entropy loss while maintaining within-class coherence. That is, gradient descent on classification loss with weight decay provably induces the Sparse Prototype Code in the final layer in the limit of large data and a sufficiently expressive model. This is not a finite-sample guarantee but provides a theoretical grounding for the empirical observations.

Experimental Findings

- **Vision (ViT, ResNet, CLIP):** Prototype alignment $A_{\text{proto}} > 0.95$ for all classes in all models. Average competing-class count $k(c) = 3$–8 out of 1000 classes (ImageNet). Competing classes match human-annotated confusable categories.
- **Audio (wav2vec):** Same geometry emerges in the audio representation space for AudioSet classification. Competing classes are acoustically similar sounds (e.g., "bark" and "growl").
- **Language (BERT, GPT-class):** Sparse Prototype Code confirmed in text classification tasks. Competing classes correspond to semantically adjacent categories (e.g., "positive" and "mildly positive" sentiment).
- **Transferability prediction:** Models with higher prototype alignment and lower competing-class count transfer better to new tasks (Pearson $r = 0.78$).

Ablations and Interpretation

Does the geometry require classification training? Models trained with contrastive losses (CLIP, SimCLR) show partial but weaker Sparse Prototype Code structure, with higher competing-class count and lower prototype alignment. Supervised classification induces stronger structure.

Does architecture matter? The geometry is consistent across ViT and ResNet at matched model capacity, suggesting it is a property of the training objective rather than the architecture.

Depth: The Sparse Prototype Code is weakest in early layers and strongest in the final representation layer, consistent with hierarchical abstraction during training.

Reference

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